WEEK 10 · SESSION 2

Spark and Distributed ETL

Transform analytical data from relational storage to distributed pipelines.

DSACADEMY

Learning outcomes

01 Explain partitions, shuffles, and lazy execution.

02 Translate a local transform to Spark.

03 Diagnose skew and avoid unnecessary movement.

The concept system

01

partitions

02

lazy evaluation

03

shuffles

04

distributed joins

A repeatable reasoning loop

01 **Define** → 02 **Transform** → 03 **Validate** → 04 **Communicate**

Scale an event aggregation from Pandas to PySpark.

Guided lab

LAB BRIEF

Scale an event aggregation from Pandas to PySpark.

workbook / tests / evidence

01 **Frame**

Define inputs, outputs, and success criteria.

02 **Build**

Implement the smallest correct workflow.

03 **Verify**

Test normal, boundary, and failure cases.

04 **Explain**

Record the evidence and remaining limits.

Portfolio deliverable

A Spark job with execution-plan notes and data checks.

QUALITY GATE

- Reproducible
- Tested
- Decision-relevant

WEEKLY ASSESSMENT

Analytical Pipeline at Two Scales

Data model 25%, SQL 25%, Spark 25%, tests 15%, documentation 10%.

SESSION CHECKPOINT

What should remain after the lesson?

CONCEPT	partitions + lazy evaluation
PRACTICE	Scale an event aggregation from Pandas to PySpark.
EVIDENCE	A Spark job with execution-plan notes and data checks.